BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: 6465

Retyped faithfully from the original board paper — same questions, no changes.

Biology - Paper: I - Objective (III)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6465

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: Objective - (III)

Q1. Scientific name of Sago-palm is:

(A) Cycas
(B) Pinus
(C) Taxus
(D) Picea

Q2. Colour of spores of smuts is:

(A) brown
(B) yellow
(C) black
(D) blue

Q3. Euglenoids are thought to be closely related to:

(A) zooflagellates
(B) dinoflagellates
(C) diatoms
(D) brown algae

Q4. Hepatitis is inflammation of:

(A) Kidney
(B) Heart
(C) Liver
(D) Lungs

Q5. Which is composed of double stranded DNA molecules?

(A) Mesosomes
(B) Ribosomes
(C) Plasmids
(D) Granules

Q6. Cell membrane is chemically composed of how much proteins?

(A) 20 - 40 %
(B) 40 - 60 %
(C) 60 - 80 %
(D) 80 - 100 %

Q7. Enzymes involved in synthesis of protein are integral part of:

(A) Mitochondria
(B) Ribosomes
(C) Chloroplast
(D) Nucleolus

Q8. Which of the given value is the heat capacity of water?

(A) 1.0
(B) 2.0
(C) 3.0
(D) 4.0

Q9. In biological control an aphid is being controlled by:

(A) Honey bee
(B) Wasp
(C) Mosquito
(D) Dragon fly

Q10. Plasmodesmata are found in:

(A) symplast
(B) apoplast
(C) protoplast
(D) chloroplast

Q11. Which constitutes about 90% of plasma?

(A) inorganic ion
(B) water
(C) proteins
(D) organic nutrients

Q12. Which help in voice production when vibrated by air?

(A) spinal cord
(B) vocal cord
(C) trachea
(D) bronchi

Q13. All of the Insectivorous plants are true.

(A) Prokaryotes
(B) Heterotrophs
(C) Autotrophs
(D) Protists

Q14. Calvin cycle is also known as:

(A) C_2 - pathway
(B) C_3 - pathway
(C) C_4 - pathway
(D) C_5 - pathway

Q15. Molecular formula of chlorophyll "a" is:

(A) $C_{55}H_{70}O_5N_4Mg$ [best-effort]
(B) $C_{55}H_{72}O_5N_4Mg$ [best-effort]
(C) $C_{55}H_{72}O_4N_4Mg$ [best-effort]
(D) $C_{55}H_{72}O_5N_4Mg$ [best-effort]

Q16. J-shaped stomach is found in:

(A) Shark
(B) Perch
(C) Trout
(D) Plaice

Q17. The larvae such as bipinnaria are found in phylum

(A) Porifera
(B) Coelentrata
(C) Echinodermata
(D) Annelida

Biology - Paper: I - Subjective

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section I is compulsory. Attempt any 3 questions from Section II.

Note: (SECTION-I)

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Draw the formation of Glycylalanine showing peptide linkage.

ii. Define Reversible and Irreversible Inhibitors.

iii. Give a diagrammatic representation of an enzyme substrate reaction (Lock and key model).

iv. Differentiate between enzyme and inhibitors.

v. Differentiate between obligate and facultative parasites.

vi. How fungi is economically helpful in the manufacture of antibiotics and other drugs?

vii. How sponges reproduce?

viii. How Platyhelminthes have adapted themselves to the parasitic mode of life?

ix. Differentiate between mantle and radula.

x. Write down three basic characteristics of Chordates.

xi. What happens to pyruvic acid before entering into citric acid cycle?

xii. Define Alcoholic Fermentation.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Differentiate between Microbiology and Biotechnology.

ii. What are endangered species? Give example.

iii. Differentiate between chromoplasts and leucoplasts.

iv. Give any two functions of Nucleolus.

v. Name a parasitic amoeba. What disease does it cause?

vi. Write two examples of zooflagellates.

vii. Give two characters of green algae similar to green plants.

viii. What are Euglenoids? Give their evolutionary significance.

ix. Define Double Fertilization. How is it important for storage of food?

x. Differentiate between homospory and heterospory.

xi. Differentiate between active and passive immunity.

xii. Differentiate between thrombus and embolus.

Q4.. (Marks 12) Write short answers to any Six parts. (6 x 2 = 12)

i. Give biological classification of Corn plant.

ii. Define Cysts.

iii. What is Swallowing?

iv. Define Dyspepsia, also give its symptoms.

v. How does Sundew get its food?

vi. Define Larynx.

vii. Define Alveoli.

viii. How do carbon dioxide and temperature affect the capacity of haemoglobin to combine with oxygen?

ix. Define Carcinoma.

> Note: (SECTION-II) (Each question carries Eight (4+4=8) Marks)

Q5.. (a) Discuss biology and the service of mankind. (b) Describe the hypothesis "Influx of K+ ions" about opening and closing of stomata.

Q6.. (a) Write a note on amino acids. (b) Discuss the importance of fungi in genetic research, food and pharmaceutical industry.

Q7.. (a) Write a note on Nutrition in Bacteria. (b) Write life cycle of an angiospermic plant.

Q8.. (a) Write a note on any four viral diseases in man. (b) Make a complete sketch of glycolysis.

Q9.. (a) Describe in detail the structure and functions of Endoplasmic Reticulum. (b) Write a note on Food poisoning.